C 21854 **(Pages : 4)** **Name.....**

Reg. No.....

SECOND SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION, APRIL 2022

Statistics

STS 2C 02—PROBABILITY DISTRIBUTIONS

(2016—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions in one word.*
Each question carries 1 *mark.*

Name the following :

1. Reciprocal of $E\left[\frac{1}{X}\right]$ of a random variable X.

2. For a random variable X, the function $E\left[e^{tX}\right]$.

3. A probability distribution with mean and standarad deviations is same.

Fill up the blanks :

4. V(X) =6, then V[2X – 4] = ————————.

5. X and Y are independent Poisson random variables with parameters 2 and 3. P(X + Y > 0) = ————————.

6. P(X > 4) when X following uniform distribution over [0,8] is ————————.

7. If X ~ N(5, 2), then P(X < 5) = ————————.

Write true *or* false :

8. Third central moment of a normal distribution is always zero.

9. The range of variation of beta distribution of first kind is from 0 to ∞.

10. Chebyshev's inequality may suggest an upper bound to the probability of an event beyond 1.

(10 × 1 = 10 marks)

Turn over

Section B

Answer **all** *questions in one sentence.*
Each question carries 2 *marks*

11. If $p(x, y) = \frac{1}{22}(2x + 3y)$, where $x = 0, 1$ and $y = 1, 2$ is a joint p.m.f., find the marginal distribution of X.

12. Find the parameters of a binomial distribution with mean 3 and variance 2.

13. Define conditional probability distribution of X/Y = y.

14. X~N(4, 1) , obtain : (i) P(X > 5) ; and (ii) P (X < 3).

15. Differentiate uniform distribution of discrete and continuous type.

16. Define Pareto distribution.

17. State Bernoulli's law of large numbers.

(7 × 2 = 14 marks)

Section C

Answer any **three** *questions.*
Each one carries 4 *marks.*

18. For a random variable X, P(X = 0) = P(X = 2) = p and P (X = 1) = 1 – 2p. Find V (X – 2).

19. State and prove multiplication theorem on expectation for two random variables X and Y.

20. Obtain the m.g.f. of X denoting the number of failures before the first success in an experiment with two possible results success and failure where the probability of success is p.

21. Write any four properties of normal distribution.

22. Using central limit theorem show that a random variable X following B (n, p) tends to $N\left(np, \sqrt{npq}\right)$ for large n.

(3 × 4 = 12 marks)

Section D

Answer any **four** *questions.*
Each one carries 6 *marks.*

23. For two random variables X and Y, the joint p.d.f.

$f(x, y) = 1 ; 0 \le x \le 1, 0 \le y \le 1$
$= 0$, otherwise

Find : (i) Cov (X, Y) ; and (ii) P(X + Y < 1).

24. State and prove Cauchy-Schwartz inequality. Use it to establish $-1 \le r_{xy} \le 1,$ where r_{xy} is Pearson's co-efficient of correlation between X and Y.

25. For rectangular distribution, with mean 6 and variance 3, obtain the m.g.f.

26. State and prove the lack of memory property of exponential distribution

27. If $X \sim N(\mu, \sigma),$ show that the interval $[\mu - 1.96\sigma, \mu + 1.96\sigma]$ contains 95% of the observations.

28. State and prove weak law of large numbers.

(4 × 6 = 24 marks)

Section E

Answer any **two** *questions.*
Each one carries 10 *marks.*

29. Let X and Y are two random variables with joint pmf $f(x, y) = x + y ; 0 \le x \le 1 ; 0 \le y \le 1.$ Find the correlation between X and Y.

30. Define Poisson distribution. Prove $\mu_{r+1} = r\lambda\mu_{r-1} + \lambda \frac{d}{d\lambda}\mu_r$ where μ_{r-1}, μ_r and μ_{r+1} are the $(r-1)^{th}$, r^{th} and $(r+1)^{th}$ central moments of X following Poisson distribution. Also show that as $\lambda \to \infty,$ Poisson distribution becomes symmetric and mesokurtic.

Turn over

31. (i) Show that the mean deviation about mean of $X \sim N(\mu, \sigma)$ is $\sqrt{\frac{2}{\pi}}\sigma$.

(ii) In a normal population 15% of the values are less than 12 and 40% are above 16.2. Find the mean and standard deviation of the population.

32. State and prove Chebysheve's Inequality. A random variable X has the p.d.f. $f(x) = e^{-x}, x > 0$.

Find the upper bound of the probability $P(|X - 1| > 2)$ using Chebysheve's inequality.

(2 × 10 = 20 marks)